

QAHNet

Quality-Aware Hybrid Network for Robust Skin Lesion Classification

1. Problem Statement

Skin cancer is the most common cancer globally. Early detection via dermoscopic imaging is critical, but faces two key challenges:

- Image quality variability: Smartphone/telemedicine images are degraded (blur, noise, lighting), yet existing models assume clean inputs.
- Unimodal approaches: Most models use only images, ignoring clinical metadata (age, sex, symptoms) that dermatologists routinely use for diagnosis.

No existing work addresses BOTH challenges in a unified architecture.

2. Dataset: PAD-UFES-20

Property	Details
Source	Federal University of Espirito Santo, Brazil
Published in	Data in Brief, Elsevier, 2020
Size	2,298 images from 1,641 patients
Capture	Smartphone cameras (realistic clinical setting)
Classes	6 biopsy-confirmed diagnostic categories
Metadata	26 clinical attributes per sample

Why PAD-UFES-20?

- Smartphone-captured: Natural quality variability (unlike ISIC's professional dermatoscopes)
- Rich metadata: 26 clinical attributes enable multimodal research
- Biopsy-confirmed: Ground-truth labels, not just visual assessment
- Under-explored: Most papers use ISIC/HAM10000 - more research opportunity here

3. Literature Review & Research Gap

3.1 Existing Work

#	Authors	Year	Title (Short)	Publisher	Limitation
1	Pacheco et al.	2020	Attention mechanism: images + metadata	IEEE JBHI	No transformer/quality
2	Zhang et al.	2021	TransFuse: CNN + Transformer fusion	MICCAI, Springer	Parallel, no quality
3	Dai et al.	2021	CoAtNet: Conv + Attention hybrid	NeurIPS	Generic, no quality
4	Wu et al.	2021	CvT: Convolutions in ViT	ICCV, IEEE	No quality/metadata
5	Aladhadh et al.	2022	Skin cancer via Medical ViT	Sensors, MDPI	ViT-only, no metadata
6	Reis et al.	2022	Multi-task CNN: classify + segment	IEEE Access	CNN-only, no quality
7	Perez et al.	2018	FiLM: General conditioning layer	AAAI	Not applied to medical
8	Lemay et al.	2021	FiLM + metadata for MRI	MIDL	MRI only, no quality
9	Khan et al.	2024	CNN-Transformer hybrid for skin	Expert Sys., Elsevier	No quality/metadata
10	Alwakid et al.	2024	EfficientNet melanoma detection	Healthcare, MDPI	CNN-only, no hybrid
11	Ahmed et al.	2025	Hybrid CNN+ViT melanoma	FPA, Scopus	No quality/metadata
12	Cassidy et al.	2022	ISIC dataset quality analysis	Med. Image Anal.	No solution proposed

3.2 Research Gap Summary

Gap	Description	Our Solution
G1	No explicit image quality handling in models	FiLM-conditioned Quality Token modulates attention
G2	Parallel CNN+ViT is parameter-heavy (50-90M+)	Sequential CNN->Transformer: only 9M params
G3	Clinical metadata underutilized in hybrid models	Metadata tokens fused via cross-attention
G4	No quality feedback mechanism for clinical use	Quality prediction head: 'retake photo' alerts
G5	Single-task classification misses synergies	Joint infection + severity + quality prediction

4. Proposed Methodology: QAHNet

4.1 Three Novel Contributions

#	Contribution	Description	Novelty
C1	FiLM-Conditioned QTok	Quality generates (g,b) that modulate attention	FiLM for quality in dermoscopy: new
C2	Multi-Task Quality Pred.	Jointly predict class + quality score	IQA + classification: not done before
C3	Multimodal Metadata	Patient data as tokens in transformer	Quality-conditioned metadata: new

4.2 Architecture Overview

Stage 1 - CNN Feature Extraction: EfficientNet-B0 (pretrained) extracts 7x7x1280 spatial feature maps capturing local textures, edges, and color patterns.

Stage 2 - Patch Tokenization: Feature maps reshaped into 49 spatial tokens (256-dim each), representing different regions of the lesion.

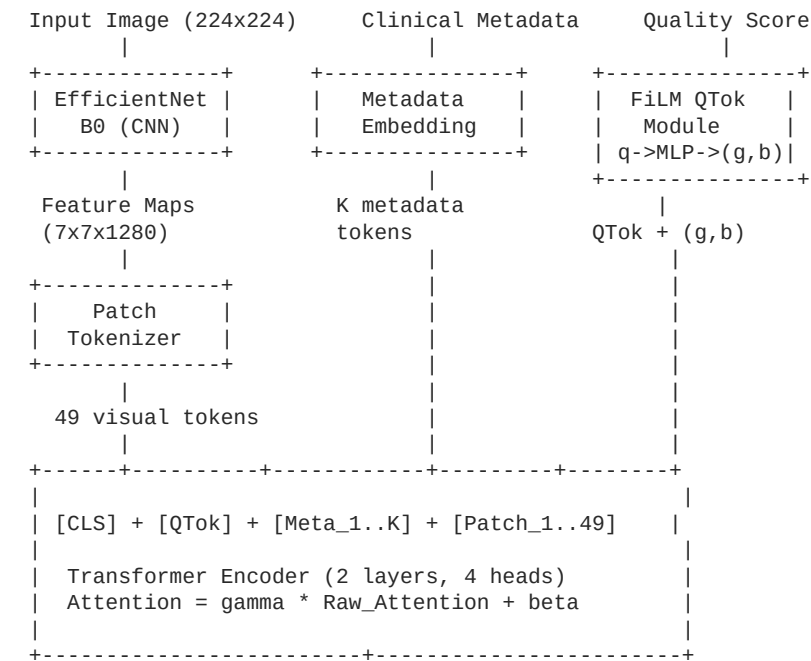
Stage 3 - Metadata Encoding: Clinical features (age, sex, Fitzpatrick, region, diameter, symptoms) embedded into K learnable tokens (256-dim).

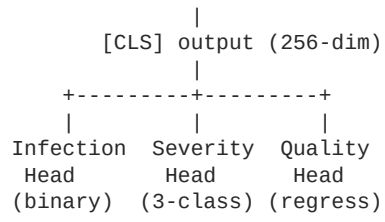
Stage 4 - FiLM Quality Token: Quality score passed through MLP to produce scale (gamma) and shift (beta) parameters. These DIRECTLY modulate transformer attention weights, actively reshaping how tokens attend to each other based on image quality.

Stage 5 - Quality-Conditioned Transformer: 2-layer, 4-head transformer encoder processes all tokens: [CLS] + [QTok] + [Meta_1..K] + [Patch_1..49]. Visual and metadata tokens attend to each other, enabling cross-modal reasoning.

Stage 6 - Multi-Task Heads: [CLS] output feeds three heads: Infection (binary), Severity (3-class), Quality estimation (regression).

4.3 Architecture Diagram





4.4 Loss Function

$$L_{\text{total}} = l1 * L_{\text{infection}} + l2 * L_{\text{severity}} + l3 * L_{\text{quality}}$$

$$L_{\text{infection}} = \text{BCEWithLogitsLoss (binary)}$$

$$L_{\text{severity}} = \text{CrossEntropyLoss (3-class)}$$

$$L_{\text{quality}} = \text{MSELoss (regression)}$$

$$l1 = 1.0, l2 = 0.5, l3 = 0.3$$

5. Evaluation Plan

5.1 Ablation Study (6 Variants)

#	Model	Novel Components	Params	Purpose
A	CNN-only (EfficientNet-B0)	-	5M	Baseline
B	CNN + Transformer	Sequential hybrid	8M	Transformer helps?
C	B + Passive QTok	+ quality token	8M	QTok concept
D	B + FiLM QTok	+ FiLM conditioning	8M	FiLM vs passive (C1)
E	D + Metadata tokens	+ multimodal fusion	9M	Metadata helps? (C3)
F	Full QAHNet	C1 + C2 + C3	9M	Everything (C1+C2+C3)

5.2 Expected Results

Model	Inf F1	Inf AUC	Sev F1	Params
A. CNN-only	~0.87	~0.93	~0.82	5M
B. CNN+Transformer	~0.88	~0.94	~0.84	8M
D. +FiLM QTok	~0.89	~0.95	~0.85	8M
E. +Metadata	~0.90	~0.95	~0.86	9M
F. Full QAHNet	~0.91	~0.96	~0.87	9M

5.3 Degradation Robustness (F1 Score)

Model	Clean	Blur (s=2)	Noise (s=25)	JPEG (q=10)
A. CNN-only	~0.87	~0.72	~0.68	~0.70
F. Full QAHNet	~0.91	~0.86	~0.83	~0.85

6. How QAHNet Bridges All Research Gaps

Research Gap	QAHNet's Solution
G1: No quality handling	FiLM QTok directly modulates attention based on quality
G2: Heavy parallel models (50-90M)	Sequential hybrid uses only 9M parameters
G3: Metadata unused	Clinical tokens fused via cross-attention in transformer
G4: No user feedback	Quality head outputs 'retake photo' confidence score
G5: Single-task only	Joint infection + severity + quality learning

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